

Program-7 : Deploy a Web Application to Kubernetes using Minikube

Description :

This program demonstrates deploying a simple Nginx web application on a local Kubernetes cluster using Minikube. The deployment is exposed as a service and accessed through a browser.

Step-1 : Start Minikube

```
minikube start
```

Step-2 : Check Minikube Status

```
minikube status
```

Step-3 : If Minikube is not running

```
minikube delete
```

```
minikube start --driver=docker
```

Step-4 : Create Project Directory

```
mkdir program-7
```

```
cd program-7
```

Step-5 : Create Deployment File

```
nano nginx-deployment.yaml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: nginx-deployment
```

```
spec:
```

```
  replicas: 2
```

```
  selector:
```

```
    matchLabels:
```

```
      app: nginx
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        app: nginx
```

```
    spec:
```

```
      containers:
```

```
        - name: nginx
```

```
          image: nginx:latest
```

```
          ports:
```

```
            - containerPort: 80
```

Step-6 : Apply the Deployment

kubectl apply -f nginx-deployment.yaml

Step-7 : Check Deployment and Pods

kubectl get deployments

kubectl get pods

Step-8 : Expose Deployment as a Service

kubectl expose deployment nginx-deployment --type=NodePort --port=80

Step-9 : Check Services

kubectl get services

Step-10 : Access the Web Application

minikube service nginx-deployment

```
apiserver: Running
kubeconfig: Configured

NAME      STATUS    ROLES    AGE      VERSION
minikube  Ready     control-plane  2m50s   v1.34.0
pavan-k@pavank1RV24MC072:~/program-7$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🔗 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📶 Pulling base image v0.0.48 ...
🔄 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass

! /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.34.0.
  ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
👉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
pavan-k@pavank1RV24MC072:~/program-7$ nano nginx-deployment.yaml
pavan-k@pavank1RV24MC072:~/program-7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
pavan-k@pavank1RV24MC072:~/program-7$ kubectl get deployments
kubectl get pods
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  1/2     2             1           15s
NAME      READY   STATUS             RESTARTS   AGE
nginx-deployment-6f9664446b-9hd6t  0/1     ContainerCreating  0           15s
nginx-deployment-6f9664446b-qt1bh  1/1     Running            0           15s
pavan-k@pavank1RV24MC072:~/program-7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
pavan-k@pavank1RV24MC072:~/program-7$ kubectl get services
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes  ClusterIP   10.96.0.1     <none>         443/TCP           6m36s
nginx-deployment  NodePort    10.109.10.87 <none>         80:31927/TCP      10s
pavan-k@pavank1RV24MC072:~/program-7$ minikube service nginx-deployment

```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31927

```
👉 Opening service default/nginx-deployment in default browser...
pavan-k@pavank1RV24MC072:~/program-7$ [30516:30516:0105/021609.936651:ERROR:chrome/browser/process_singleton_posix.cc:358] The profile appears to be in use by another Google Chrome process (
9004) on another computer (pavan-k-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ). Chrome has locked the profile so that it doesn't get corrupted. If you are sure no other processes are using th
is profile, you can unlock the profile and relaunch Chrome.
[30516:30516:0105/021609.939223:ERROR:chrome/browser/ui/views/message_box_dialog.cc:199] Unable to show message box: Google Chrome - The profile appears to be in use by another Google Chrome
process (9004) on another computer (pavan-k-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ). Chrome has locked the profile so that it doesn't get corrupted. If you are sure no other processes ar
e using this profile, you can unlock the profile and relaunch Chrome.

```

Step-11 : Cleanup Pods

kubectl delete pods --all

Step-12 : Delete All Services

kubectl delete svc --all

Step-13 : Delete All Deployments

kubectl delete deployments --all